NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2024-2025

IE2106 – ENGINEERING MATHEMATICS I

April / May 2025 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 13 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.
6. A list of useful formulae is given in Appendix A on pages 5-6, followed by Statistical Tables on pages 7-11, and the Laplace Transform Table on pages 12-13.

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1. (a) The number of flash floods that happen in Bedok over a period of time can be modelled by a Poisson distribution. On the average, Bedok experiences 4 flash floods in a year.

   (i) What is the probability that at least 2 flash floods happen in Bedok in a quarter? Give your answer correct to 3 decimal places.

   (ii) Since April 2025, Bedok has already experienced 2 flash floods. What is the probability that Bedok will experience at most 3 flash floods in the second quarter of 2025? Give your answer correct to 3 decimal places.

   (9 Marks)

Note: Question No. 1 continues on page 2.

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(b) Singapore receives a lot of rain, the general perception is that the amount of rain is increasing over the years. A scientist is studying the climate change effect on the rainfall in Singapore. A sample of 61 monthly rainfall data spanning over the past few years is used for the study. The sample has an average monthly rainfall of 195 mm and standard deviation of 12 mm.

(i) Based on the sample, find the 97% confidence interval for the average monthly rainfall in Singapore. State any assumption needed. Explain the meaning of 97% confidence interval.

(ii) The standard deviation of monthly rainfalls in 1950s is known to be 10 mm. From the sample used by the scientist, can the scientist draw the conclusion that the standard deviation of monthly rainfalls has increased? Test with $\alpha = 5\%$. State any assumption needed. What is the possible range of α to reverse the hypothesis test conclusion?

(16 Marks)

2. (a) Let X be a random variable with the following probability density function

$$f(x) = \begin{cases} c, & 0 \le x \le 1 \\ c\,x^{1/4}, & 1 \le x \le 16 \\ 0, & \text{otherwise} \end{cases}$$

where c is a constant. Find c. Hence, compute $E(X^{3/4})$, $Var(X^{3/4})$, $E(1-X^{3/4})$ and $Var(2X^{3/4}-5)$, give your answers correct to 3 decimal places.

(11 Marks)

(b) The following points are given:

x	-1	2	3
$g(x)$	2	0	1

(i) Obtain the Newton's divided difference interpolating polynomial for the given points (no need to simplify the polynomial). Express all your answers in fractions.

(ii) Suppose a new point $(x, g(x)) = (4, 3/2)$ is added to the above table. Show how you can build on part (i) to get the new Newton's divided difference interpolating polynomial. Hence, estimate the value of $g(3.5)$.

(14 Marks)

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3. (a) (i) Find the Laplace transform of the following function

$$f(t) = e^{-t}[\sin(t-1) + \cos(t-1)]u(t-1),$$

where $u(t)$ is the unit step function.

(ii) Find the inverse Laplace transform of the following function

$$F(s) = \frac{2s^2 - 6s + 13}{s(s^2 - 4s + 13)}$$

(12 Marks)

(b) A linear system is described by the following differential equation:

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = r(t),$$

where $r(t)$ is the input and $y(t)$ is the output. Using the method of Laplace transform, find the output of the system due to the input $r(t) = 2e^{-t}u(t)$ under the initial conditions:

$$y(0) = 0, \frac{dy(0)}{dt} = 1,$$

where $u(t)$ is the unit step function.

(13 Marks)

4. (a) Consider the following partial differential equation:

$$\frac{\partial^2 y}{\partial x \partial t} = te^{-x} + e^{-t}.$$

(i) Find the general solution of the partial differential equation.
(ii) Given the boundary and initial conditions:

$$y(0,t) = \frac{1}{2}t^2, \; y(x,0) = -x,$$

find the solution of the partial differential equation.

(12 Marks)

Note: Question No. 4 continues on page 4.

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(b) Consider the partial differential equation:

$$\frac{\partial y(x,t)}{\partial x} + 2\frac{\partial y(x,t)}{\partial t} = 0$$

(i) Solve for $y(x,t)$ using the method of separation of variables.

(ii) Solve for $y(x,t)$ using the method of Laplace transform, assuming the boundary and initial conditions:

$$y(0,t) = u(t);\ y(x,0) = 0,\ \text{for any}\ x > 0,$$

where $u(t)$ is the unit step function.

(13 Marks)

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Appendix A

Total Probability of A

$$P(A) = \sum_{j=1}^{k} P(A \mid B_j) P(B_j)$$

Bayes Theorem

$$P(B_r \mid A) = \frac{P(A \mid B_r) P(B_r)}{\sum_{j=1}^{k} P(A \mid B_j) P(B_j)}$$

Poisson Distribution

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

Binomial Distribution

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Fixed-Point Iteration Method

$$x_{n+1} = g(x_n), \quad n = 0, 1, 2, \ldots$$

Euler Method

$$y_{n+1} = y_n + hf(x_n, y_n)$$

Improved Euler Method (Heun's Method)

$$a = hf(x_n, y_n)$$
$$b = hf(x_{n+1}, y_n + a)$$
$$y_{n+1} = y_n + \frac{1}{2}(a + b)$$

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Appendix A (Continued)

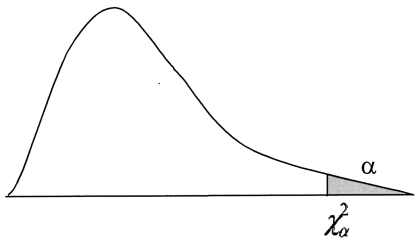
Runge-Kutta Method

$$a = h\,f(x_n, y_n)$$
$$b = h\,f(x_n + \tfrac{1}{2}h,\ y_n + \tfrac{1}{2}a)$$
$$c = h\,f(x_n + \tfrac{1}{2}h,\ y_n + \tfrac{1}{2}b)$$
$$d = h\,f(x_n + h,\ y_n + c)$$
$$y_{n+1} = y_n + \tfrac{1}{6}(a + 2b + 2c + d)$$

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Statistical Table - 1

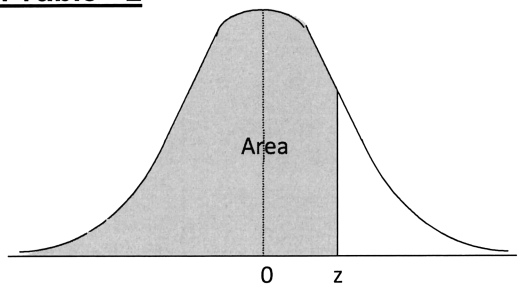
Percentage Points of χ^2 Distribution



d.f. \ α	.99	.975	.95	.90	.50	.10	.05	.025	.01
1	.0002	.001	.004	.02	.45	2.71	3.84	5.02	6.63
2	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21
3	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34
4	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28
5	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09
6	.87	1.24	1.64	2.20	5.35	10.64	12.59	14.45	16.81
7	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48
8	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09
9	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67
10	2.56	3.24	3.94	4.87	9.34	15.99	18.31	20.48	23.21
11	3.05	3.81	4.57	5.58	10.34	17.28	19.68	21.92	24.72
12	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22
13	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69
14	4.66	5.62	6.57	7.79	13.34	21.06	23.68	26.12	29.14
15	5.23	6.26	7.26	8.55	14.34	22.31	25.00	27.49	30.58
16	5.81	6.90	7.96	9.31	15.34	23.54	26.30	28.85	32.00
17	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41
18	7.01	8.23	9.39	10.86	17.34	25.99	28.87	31.53	34.81
19	7.63	8.90	10.12	11.65	18.34	27.20	30.14	32.85	36.19
20	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57
21	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93
22	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29
23	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64
24	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98
25	11.52	13.11	14.61	16.47	24.34	34.38	37.65	40.65	44.31
26	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64
27	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96
28	13.56	15.30	16.93	18.94	27.34	37.92	41.34	44.46	48.28
29	14.26	16.04	17.71	19.77	28.34	39.09	42.56	45.72	49.59
30	14.95	16.78	18.49	20.60	29.34	40.26	43.77	46.98	50.89
40	22.16	24.42	26.51	29.05	39.34	51.81	55.76	59.34	63.69
50	29.71	32.35	34.76	37.69	49.33	63.17	67.50	71.42	76.15
60	37.48	40.47	43.19	46.46	59.33	74.40	79.08	83.30	88.38
70	45.44	48.75	51.74	55.33	69.33	85.53	90.53	95.02	100.43
80	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33
90	61.75	65.64	69.13	73.29	89.33	107.57	113.15	118.14	124.12
100	70.06	74.22	77.93	82.36	99.33	118.50	124.34	129.56	135.81

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Statistical Table - 2



Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0352	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0722	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Note: Statistical Table 2 continues on page 9.

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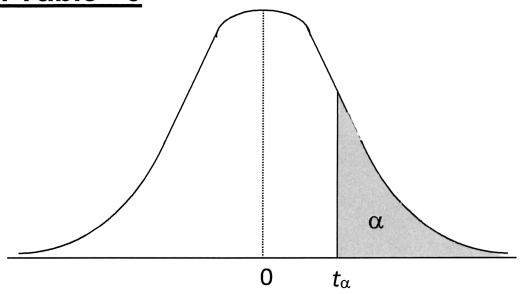
Statistical Table – 2 (Continued)

Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9278	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

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Statistical Table - 3



Critical Values of the *t*-Distribution

	α						
ν	0.40	0.30	0.20	0.15	0.10	0.05	0.025
1	0.325	0.727	1.376	1.963	3.078	6.314	12.706
2	0.289	0.617	1.061	1.386	1.886	2.920	4.303
3	0.277	0.584	0.978	1.250	1.638	2.353	3.182
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228
11	0.260	0.540	0.876	1.088	1.363	1.796	2.201
12	0.259	0.539	0.873	1.083	1.356	1.782	2.179
13	0.259	0.537	0.870	1.079	1.350	1.771	2.160
14	0.258	0.537	0.868	1.076	1.345	1.761	2.145
15	0.258	0.536	0.866	1.074	1.341	1.753	2.131
16	0.258	0.535	0.865	1.071	1.337	1.746	2.120
17	0.257	0.534	0.863	1.069	1.333	1.740	2.110
18	0.257	0.534	0.862	1.067	1.330	1.734	2.101
19	0.257	0.533	0.861	1.066	1.328	1.729	2.093
20	0.257	0.533	0.860	1.064	1.325	1.725	2.086
21	0.257	0.532	0.859	1.063	1.323	1.721	2.080
22	0.256	0.532	0.858	1.061	1.321	1.717	2.074
23	0.256	0.532	0.858	1.060	1.319	1.714	2.069
24	0.256	0.531	0.857	1.059	1.318	1.711	2.064
25	0.256	0.531	0.856	1.058	1.316	1.708	2.060
26	0.256	0.531	0.856	1.058	1.315	1.706	2.056
27	0.256	0.531	0.855	1.057	1.314	1.703	2.052
28	0.256	0.530	0.855	1.056	1.313	1.701	2.048
29	0.256	0.530	0.854	1.055	1.311	1.699	2.045
30	0.256	0.530	0.854	1.055	1.310	1.697	2.042
40	0.255	0.529	0.851	1.050	1.303	1.684	2.021
60	0.254	0.527	0.848	1.045	1.296	1.671	2.000
120	0.254	0.526	0.845	1.041	1.289	1.658	1.980
∞	0.253	0.524	0.842	1.036	1.282	1.645	1.960

Note: Statistical Table 3 continues on page 11.

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Statistical Table - 3 (Continued)

Critical Values of the *t*-Distribution

	α						
ν	0.02	0.015	0.01	0.0075	0.005	0.0025	0.0005
1	15.895	21.205	31.821	42.434	63.657	127.322	636.590
2	4.849	5.643	6.965	8.073	9.925	14.089	31.598
3	3.482	3.896	4.541	5.047	5.841	7.453	12.924
4	2.999	3.298	3.747	4.088	4.604	5.598	8.610
5	2.757	3.003	3.365	3.634	4.032	4.773	6.869
6	2.612	2.829	3.143	3.372	3.707	4.317	5.959
7	2.517	2.715	2.998	3.203	3.499	4.029	5.408
8	2.449	2.634	2.896	3.085	3.355	3.833	5.041
9	2.398	2.574	2.821	2.998	3.250	3.690	4.781
10	2.359	2.527	2.764	2.932	3.169	3.581	4.587
11	2.328	2.491	2.718	2.879	3.106	3.497	4.437
12	2.303	2.461	2.681	2.836	3.055	3.428	4.318
13	2.282	2.436	2.650	2.801	3.012	3.372	4.221
14	2.264	2.415	2.624	2.771	2.977	3.326	4.140
15	2.249	2.397	2.602	2.746	2.947	3.286	4.073
16	2.235	2.382	2.583	2.724	2.921	3.252	4.015
17	2.224	2.368	2.567	2.706	2.898	3.222	3.965
18	2.214	2.356	2.552	2.689	2.878	3.197	3.922
19	2.205	2.346	2.539	2.674	2.861	3.174	3.883
20	2.197	2.336	2.528	2.661	2.845	3.153	3.849
21	2.189	2.328	2.518	2.649	2.831	3.135	3.819
22	2.183	2.320	2.508	2.639	2.819	3.119	3.792
23	2.177	2.313	2.500	2.629	2.807	3.104	3.768
24	2.172	2.307	2.492	2.620	2.797	3.091	3.745
25	2.167	2.301	2.485	2.612	2.787	3.078	3.725
26	2.162	2.296	2.479	2.605	2.779	3.067	3.707
27	2.158	2.291	2.473	2.598	2.771	3.057	3.690
28	2.154	2.286	2.467	2.592	2.763	3.047	3.674
29	2.150	2.282	2.462	2.586	2.756	3.038	3.659
30	2.147	2.278	2.457	2.581	2.750	3.030	3.646
40	2.125	2.250	2.423	2.542	2.704	2.971	3.551
60	2.099	2.223	2.390	2.504	2.660	2.915	3.460
120	2.076	2.196	2.358	2.468	2.617	2.860	3.373
∞	2.054	2.170	2.326	2.432	2.576	2.807	3.291

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Laplace Transform Table

		$f(t)$	$F(s)$
1.	Unit impulse	$\delta(t)$	1
2.	Unit step	$u(t)$	$\frac{1}{s}$
3.	Unit ramp	$r(t) = tu(t)$	$\frac{1}{s^2}$
4.	Unit parabola	$p(t) = \frac{1}{2}t^2u(t)$	$\frac{1}{s^3}$
5.	Exponential	e^{-at}	$\frac{1}{s+a}$
6.	t-Multiplication exponential	te^{-at}	$\frac{1}{(s+a)^2}$
7.	Sine	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
8.	Cosine	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
9.	Damped Sine	$e^{-at}\sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
10.	Damped Cosine	$e^{-at}\cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
11.	Linearity	$c_1f_1(t)+c_2f_2(t)$	$c_1F_1(s)+c_2F_2(s)$
12.	Differentiation	$\frac{d}{dt}f(t)$	$sF(s)-f(0)$
13.	n-Fold differentiation	$\frac{d^n}{dt^n}f(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)$ $-\cdots\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$
14.	Integration	$\int_{0-}^{t} f(\tau)d\tau$	$\frac{F(s)}{s}$
15.	t-Multiplication	$tf(t)$	$\frac{-d}{ds}F(s)$

Note: Laplace Transform Table continues on page 13.

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Laplace Transform Table (Continued)

	$f(t)$	$F(s)$
16. n-Fold t-Multiplication	$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n} F(s)$
17. Time shift	$f(t-t_0)u(t-t_0)$; $t_0 > 0$	$e^{-st_0} F(s)$
18. Frequency shift	$e^{-s_0 t} f(t)$	$F(s+s_0)$
19. Time-frequency scaling	$f(ct)$; $c > 0$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
20. Periodic Function	$f(t) = f_1(t)u(t) + f_1(t-T)u(t-T) + f_1(t-2T)u(t-2T) + \cdots$	$\frac{1}{1-e^{-sT}} F_1(s)$

END OF PAPER

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IE2106 ENGINEERING MATHEMATICS I

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